

# Ju Peng (彭举) | Curriculum Vitae

**Affiliation:** School of Geospatial Artificial Intelligence, East China Normal University, Shanghai, China

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**Research Interests:** Spatiotemporal data mining and analysis, Human mobility, GeoAI



## Work Experience

### East Normal China University (ENCU)

07/2026-present

Postdoctoral fellow in School of Geospatial Artificial Intelligence

Supervisor: Prof. Jonathan Li

## Education

### Central South University (CSU)

09/2021-06/2026

Ph.D. in Cartography and Geographic Information Engineering

Supervisors: Prof. Min Deng & Prof. Jianbo Tang

Honors: President's Scholarship for Innovation (Multiple), First-Class Academic Scholarship (Multiple), CSU Outstanding Student

### Central South University (CSU)

09/2017-06/2021

B.Sc. in Geographic Information Science

Ranking: 3/54 | Selected for "Undergraduate-Postgraduate Integrated Innovation Program"

Honors: National Scholarship, National Encouragement Scholarship, Outstanding Graduate of Hunan Province

## Publications and Patents

- **Ju Peng**, Jianbo Tang\*, Min Deng, *et al.* A two-stage method for detecting trajectory clusters of different densities with peak trajectories identification[J]. *International Journal of Geographical Information Science*, 2025: 1-34. <https://doi.org/10.1080/13658816.2025.2478463>. (SCI, JCR Q1)
- **Ju Peng**, Min Deng, Jianbo Tang\*, *et al.* A movement-aware measure for trajectory similarity and its application for ride-sharing path extraction in a road network[J]. *International Journal of Geographical Information Science*, 2024, 38(9): 1703-1727. <https://doi.org/10.1080/13658816.2024.235369>. (SCI, JCR Q1)
- Jianbo Tang, Min Deng, **Ju Peng**\*, *et al.* How to describe spatiotemporal patterns of moving objects: A new classification framework[J]. *International Journal of Geographical Information Science*, 2026, 40(4): 1005-1046. <https://doi.org/10.1080/13658816.2025.2548819> (SCI, JCR Q1)
- Jianbo Tang, Min Deng, **Ju Peng**\*, *et al.* Automatic road network selection method considering functional semantic features of roads with graph convolutional networks [J]. *International Journal of Geographical Information Science*, 2024, 38(11): 2403-2432. <https://doi.org/10.1080/13658816.2024.238719> (SCI, JCR Q1)
- Min Deng (advisor), **Ju Peng**, Jianbo Tang\*, *et al.* Spatio-temporal Pattern Mining of Moving Objects: Concepts, Methods, and Challenges [J]. *Journal of Geovisualization and Spatial Analysis*, 2025, 9(2):1-32. <https://doi.org/10.1007/s41651-025-00236-1> (ESCI, JCR Q1)
- **Ju Peng**, Jianbo Tang \*, Min Deng, *et al.* Discovering spatiotemporal patterns of human outdoor activities with crowdsourced trajectory data[J]. *Geo-spatial Information Science*, 2025, 28(3): 1214-1236. <https://doi.org/10.1080/10095020.2025.2506772> (SCI, JCR Q1)
- **Ju Peng**, Huimin Liu, Jianbo Tang\*, *et al.* Exploring crowd travel demands based on the characteristics of spatiotemporal

interaction between urban functional zones[J]. *ISPRS International Journal of Geo-Information*, 2023, 12(6): 225. <https://doi.org/10.3390/ijgi12060225> (SCI, JCR Q2)

- Jianbo Tang, Yuxin Zhao, Xuexi Yang\*, Min Deng, Huimin Liu, Chen Ding, **Ju Peng**, Xiaoming Mei. Statistical and density-based clustering of geographical flows for crowd movement patterns recognition[J]. *Applied Soft Computing*, 2024, 163: 111912. <https://doi.org/10.1016/j.asoc.2024.111912> (SCI, JCR Q1)
- Jingyi Liu, Zhiyuan Hu\*, Jianbo Tang, **Ju Peng**, Qi Guo, Xinyu Pei. STACS: a spatiotemporal adaptive clustering–segmentation algorithm for fishing activity recognition [J]. *Applied Science*, 2025, 15(16): 9107. (SCI, JCR Q2)
- Jianbo Tang, Zhiyuan Hu, **Ju Peng\***, *et al.* Road intersection identification from crowdsourced trajectory data by integrating visual and movement features. *Acta Geodaetica et Cartographica Sinica*, 2025, 54(01): 1001-1595. <http://dx.doi.org/10.11947/j.AGCS.2025.20240101> (EI, T1 Journal of China Geographic Society)
- Jingyi Liu, **Ju Peng\***, Jianbo Tang, *et al.* An adaptive trajectory clustering method integrating multiple features. *Journal of Geo-Information Science*, 2023, 25(07): 1363-1377. <http://dx.doi.org/10.12082/dqxkx.2023.230066> (EI, T2 Journal of China Geographic Society)
- Jianbo Tang\*, Heyan Xia, **Ju Peng**, *et al.* Constructing hiking navigation road network maps using crowdsourced trajectory data. *Journal of Geo-Information Science*, 2025, 27(01): 151-166. <http://dx.doi.org/10.12082/dqxkx.2025.240479> (EI, T2 Journal of China Geographic Society)
- **Ju Peng**, Jianbo Tang, Zhiyuan Hu, *et al.* A method and device for mining movement patterns of moving targets based on trajectories. Chinese Patent: ZL202510216690.2 (Granted 2025-05-16)
- **Ju Peng**, Jianbo Tang, Zhiyuan Hu, *et al.* A trajectory clustering method, apparatus, device, and medium based on peak trajectories. Chinese Patent: ZL202311636437.X (Granted 2024-03-19)
- Jianbo Tang, **Ju Peng**, Min Deng, *et al.* Multi-source data collaborative method for constructing human-vehicle-road-network integration. Chinese Patent: ZL202210909277.0 (Granted 2024-07-30)
- Jianbo Tang, **Ju Peng**, Zhiyuan Hu, *et al.* A method for generating outdoor 3D pedestrian navigation road network maps integrating crowdsourced trajectory data. Chinese Patent: ZL202311461668.1 (Granted 2024-03-19)
- Jianbo Tang, **Ju Peng**, Jingyi Liu, *et al.* A trajectory simulation method for behavioral pattern recognition of moving targets. Chinese Patent Application: CN202310600163.2 (Substantive examination)

## Research Projects

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- **Principal Investigator (PI), Hunan Provincial Graduate Research Innovation Project:** Construction and intelligent recognition of movement patterns and behavior semantics from moving objects trajectories (No. CX20220193, 2022–2024)  
I was responsible for project application, design, implementation, and publications.
- **Principal Investigator (PI), CSU Graduate Independent Exploration Innovation Project:** Same as above (No. 2022zzts105, 2022–2024)
- **Participant, National Key R&D Program Sub-project of China:** Spatiotemporal information extraction and alignment fusion (No. 2022YFB3904203, 2022–2025)  
I focused on trajectory data information extraction and alignment with other modalities.
- **Participant, NSFC General Project:** Crowd-sourced collaborative perception models for dynamic updating of high-precision urban road network maps (No. 42271462, 2022–2026)  
I was responsible for developing multi-feature intelligent road intersection recognition methods.
- **Core members, National Key R&D Sub-project of China:** Multi-source spatiotemporal data warehouse organization (No. 2024YFB3909601, 2025–2028)  
I was responsible for the design of spatiotemporal cube representation models and spatiotemporal semantic extraction from trajectory data.
- **Core members, 54th Research Institute of CETC Open Fund Project:** Semantic system construction and intelligent recognition of multi-domain combat movement behaviors (2022–2023)

I was responsible for constructing the movement behavior framework and simulation datasets for city, sea, and air targets.

- **Core members, CSU Design and Survey Institute Enterprise Project:** Sand and gravel supply-demand balance and dynamic dispatch system (2021)  
I contributed to the design of optimization algorithms and middleware procurement.

## Awards and Honors

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- **Outstanding Graduate** of CSU (2026)
- **"Emerging GIS Star"** at the 13th GIS Forum for Higher Education Institutions (2025)
- **"Best Presentation Award"** at the 11th National Doctoral Student Forum on Geographic Information Science (2024)
- **CSU President's Innovation Scholarship** (2024 & 2025)
- **Outstanding Student** of CSU (2024)
- **Second Prize** in the 18th China Graduate Mathematical Contest in Modeling (2021)
- **Outstanding Graduate** of Hunan Province & Outstanding Graduate of CSU (2021)
- **National Scholarship for Undergraduates**, CSU (2020)
- **Outstanding Student Leader**, CSU (2019)
- **Special Prize** in the 11th National University Student Surveying and Mapping Technology Paper Competition ("South Surveying and Mapping Cup") (2019)

## Academic Exchange

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- 13th University GIS Forum for Higher Education Institutions | Shanghai, China, 2025.11.07-09 (Oral Presentation)
- 2024 AAG Annual Meeting | Honolulu, USA, 2024.04.16-20 (Oral Presentation)
- 11th National Cartography and GIS Conference | Nanjing, China, 2025.09.19-21 (Oral Presentation)
- 11th GIS Doctoral Student Forum on Geographic Information Science | Nanjing, China, 2024.11.29-12.01 (Oral Presentation)
- 19th GIS Theory and Methods Annual Meeting | Xi'an, China, 2024.10.19-20 (Oral Presentation)
- 7th Cartography Theory and Methods Seminar | Changsha, China, 2024.11.15-17 (Attendee)
- 18th GIS Theory and Methods Annual Meeting | Guilin, China, 2023 (Attendee)
- 9th University GIS Forum for Higher Education Institutions | Changsha, China, 2021.10.29-31 (Attendee)
- 10th Global GIS Developer Conference (WGDC) | Changsha, China, 2021.05.18-19 (Attendee)

## Skills

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### Professional Skills

- Systematic expertise in spatiotemporal data processing, mining and analysis, especially movement pattern recognition of moving objects.
- Proficient in cartography, geospatial data processing software, spatiotemporal clustering, statistical analysis, pattern recognition, and visualization for geographic data analytics.

### Computer Skills

- Proficient in Python (algorithm design & system development), ArcGIS, QGIS.
- During PhD, independently developed a prototype system for movement pattern mining and analysis of moving objects.
- Skilled in Office, Photoshop, etc.

### Language Skills

- CET-6 certified. English.
- Strong capabilities in academic writing and presentation.